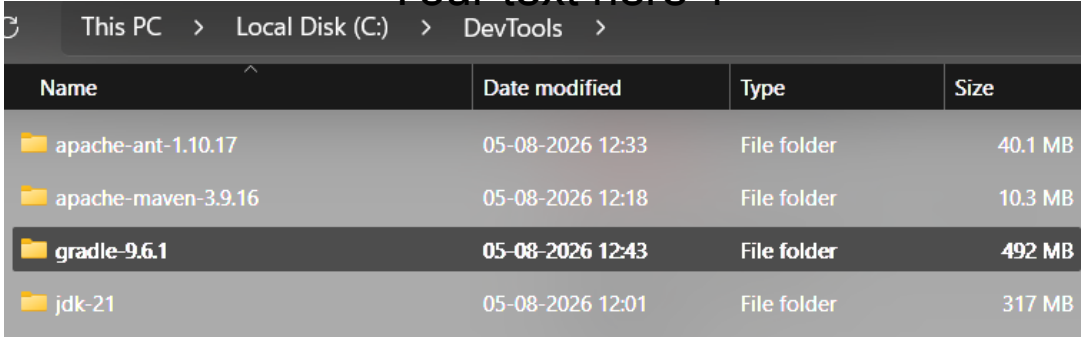


Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Prof. Snehal Andhare

Student Name	Pratik Pujari
Roll Number	23102B0008
Grade and Subject Teacher's Signature	

Experiment Number	03	
Experiment Title	To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to set up a build Job.	
Resources / Apparatus Required	Hardware: Computer system	Software: Windows, Jenkins, Any IDE with terminal
	Dev Tools folder structure: Your text here 1 	
	verify-tools.bat output:	

```
java -version
java version "21.0.11" 2026-04-21 LTS
Java(TM) SE Runtime Environment (build 21.0.11+9-LTS-211)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.11+9-LTS-211, mixed mode, sharing)

javac -version
javac 21.0.11

git --version
git version 2.45.1.windows.1

mvn -version
Apache Maven 3.9.16 (2bddd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: C:\DevTools\apache-maven-3.9.16
Java version: 21.0.11, vendor: Oracle Corporation, runtime: C:\DevTools\jdk-21
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

ant -version
Apache Ant(TM) version 1.10.17 compiled on April 6 2026

gradle -version

-----
Gradle 9.6.1
-----

Build time:    2026-06-26 14:25:50 UTC
Revision:     309d128bd9fe8c0b71311878fc660b9cbaa07c51

Kotlin:       2.3.21
Groovy:       4.0.32
Ant:          Apache Ant(TM) version 1.10.17 compiled on April 6 2026
Launcher JVM: 21.0.11 (Oracle Corporation 21.0.11+9-LTS-211)
Daemon JVM:   C:\DevTools\jdk-21 (no Daemon JVM specified, using current Java home)
OS:           Windows 11 10.0 amd64

Press any key to continue . . . |
```

Testing Maven Build

```
[INFO] Building jar: C:\Projects\Devops\ci-jenkins-multi-build-demo\target\ci-jenkins-demo-1.0.0.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 26.292 s
[INFO] Finished at: 2026-08-08T22:54:30+05:30
[INFO] -----
PS C:\Projects\Devops\ci-jenkins-multi-build-demo>
```

Running the build

```
● PS C:\Projects\Devops\ci-jenkins-multi-build-demo> java -cp target\ci-jenkins-demo-1.0.0.jar ci-jenkins-demo
Continuous Integration demo is running.
2 + 3 = 5
```

Testing Ant Build

```
test:
[junit] Running com.vit.demo.AppTest
[junit] Testsuite: com.vit.demo.AppTest
[junit] TEST PASSED: App.add(2, 3) returned 5.
[junit] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 sec
[junit] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 sec
[junit] ----- Standard Output -----
[junit] TEST PASSED: App.add(2, 3) returned 5.
[junit] -----
[junit]
[junit]
[junit] Testcase: testAdd took 0.006 sec

jar:
[mkdir] Created dir: C:\Projects\Devops\ci-jenkins-multi-build-demo\ant-build\dist
[jar] Building jar: C:\Projects\Devops\ci-jenkins-multi-build-demo\ant-build\dist\ci-jenkins-demo.jar

build:
[echo] ANT BUILD SUCCESSFUL

BUILD SUCCESSFUL
Total time: 2 seconds
```

Running the build

```
PS C:\Projects\Devops\ci-jenkins-multi-build-demo> java -jar ant-build\dist\ci-jenkins-demo.jar
Continuous Integration demo is running.
2 + 3 = 5
```

Testing Gradle Build

```
PS C:\Projects\Devops\ci-jenkins-multi-build-demo> gradle clean build

> Task :test

AppTest > testAdd STANDARD_OUT
    TEST PASSED: App.add(2, 3) returned 5.

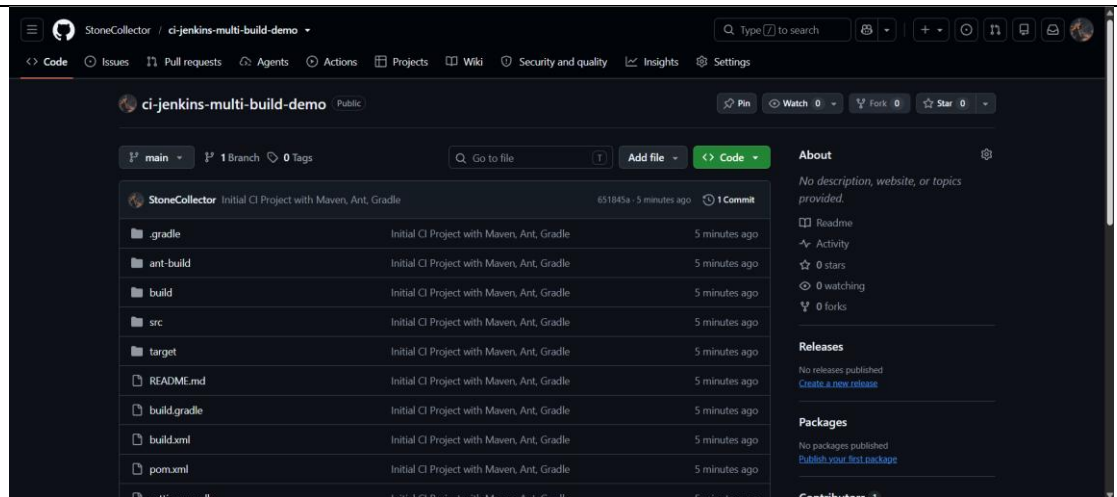
AppTest > testAdd PASSED

BUILD SUCCESSFUL in 2s
5 actionable tasks: 5 executed
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/7.4/userguide/configuration_cache_enabling.html
PS C:\Projects\Devops\ci-jenkins-multi-build-demo>
```

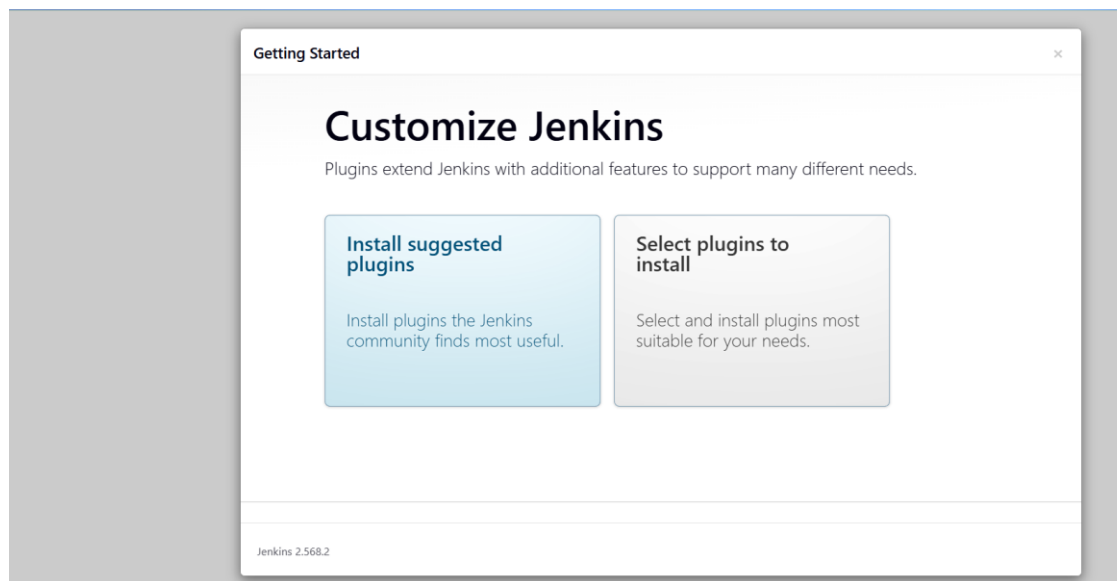
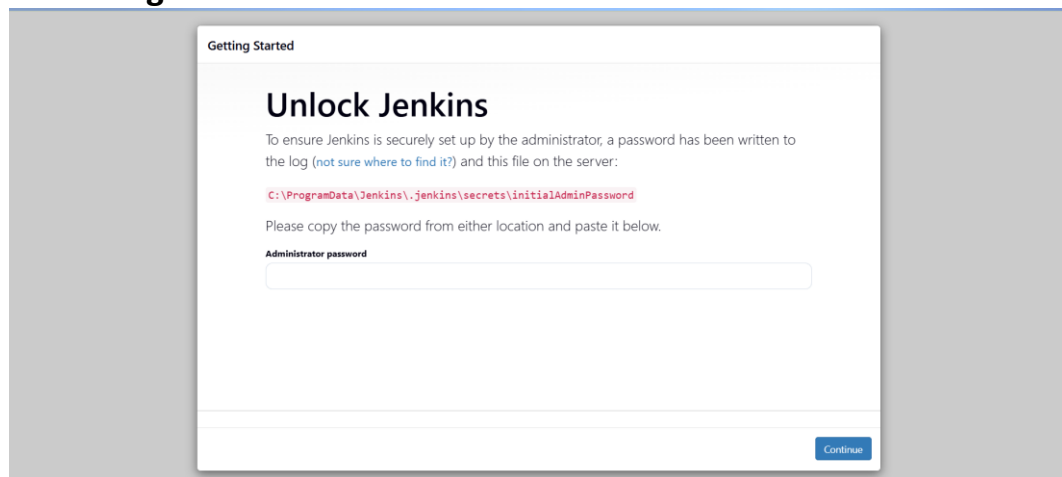
Running the build

```
PS C:\Projects\Devops\ci-jenkins-multi-build-demo> java -cp build\libs\ci-jenkins-demo-1.0.0.jar
Continuous Integration demo is running.
2 + 3 = 5
```

Git Repo Initialization



Unlocking Jenkins



Maven Build in Jenkins

Jenkins / All / New Item

New Item

Enter an item name

CI-Maven-Build

Select an item type

- Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Jenkins / CI-Maven-Build / Configure

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Repository URL ?

https://github.com/StoneCollector/ci-jenkins-multi-build-demo.git

Credentials ?

- none - + Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

Save Apply

Jenkins / CI-Maven-Build / #1

```
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.26/plexus-interpolation-1.26.jar (85 kB at 449 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.9/xz-1.9.jar (136 kB at 535 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.jar (211 kB at 920 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar (267 kB at 800 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.0/maven-archiver-3.6.0.jar (26 kB at 74 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.21/commons-compress-1.21.jar (1.0 MB at 2.6 MB/s)
[INFO] Building jar: C:\ProgramData\Jenkins\workspace\CI-Maven-Build\target\ci-jenkins-demo-1.0.0.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 32.327 s
[INFO] Finished at: 2026-08-08T23:58:14+05:30
[INFO] -----
Archiving artifacts
Finished: SUCCESS
```

Jenkins 2.568.2

CI-Maven-Build

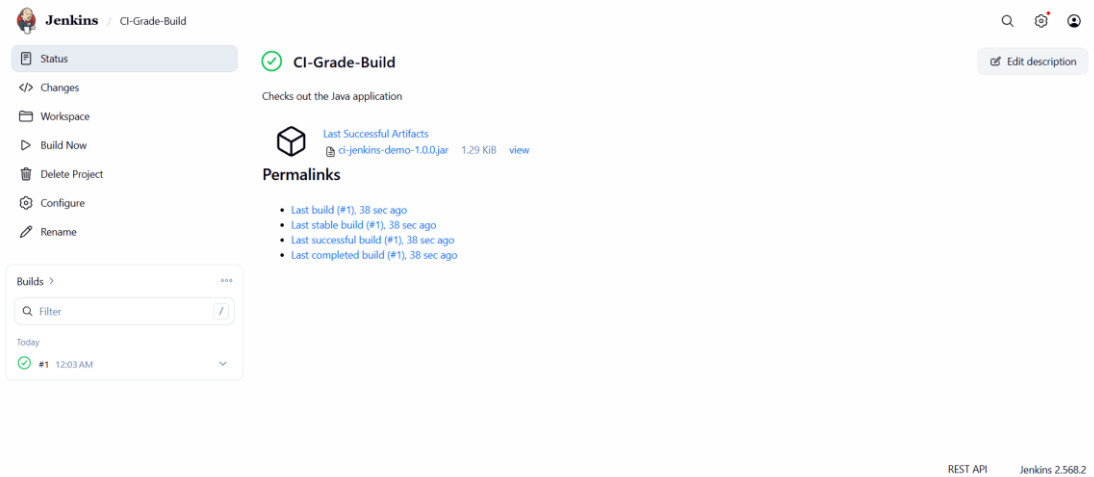
Checks out the Java application



Last Successful Artifacts

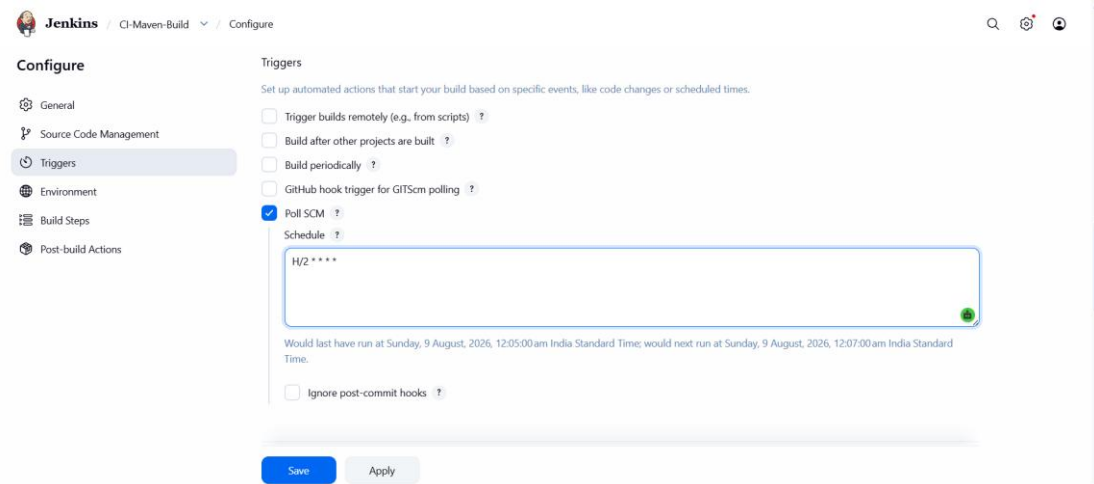
ci-jenkins-demo-1.0.0.jar 2.60 KiB [view](#)

Ant Build in Jenkins



The screenshot shows the Jenkins interface for a job named 'CI-Grade-Build'. The left sidebar contains navigation links: Status, Changes, Workspace, Build Now, Delete Project, Configure, and Rename. The main area displays the job's status as 'Checks out the Java application' with a green checkmark. Below this, it shows 'Last Successful Artifacts' as 'ci-jenkins-demo-1.0.0.jar' (1.29 KiB) with a 'view' link. A 'Permalinks' section lists four links for the last build, last stable build, last successful build, and last completed build, all indicating they occurred 38 seconds ago. A 'Builds' section shows a single build (#1) at 12:03 AM. The bottom right corner displays 'REST API' and 'Jenkins 2.568.2'.

Build Automation For maven



The screenshot shows the Jenkins configuration page for a job named 'CI-Maven-Build'. The left sidebar contains navigation links: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'Triggers' section is selected, showing options to set up automated actions. The 'Poll SCM' option is checked. The 'Schedule' field contains the text 'H/2 * * * *'. Below the schedule field, it indicates the last run was on Sunday, 9 August, 2026, at 12:05:00am India Standard Time, and the next run will be on Sunday, 9 August, 2026, at 12:07:00am India Standard Time. There is an 'Ignore post-commit hooks' checkbox. At the bottom, there are 'Save' and 'Apply' buttons.

Testing with modifications

ci-jenkins-multi-build-demo > src > main > java > com > vit > demo > App.java

```
1 0;  
2  
3  
4 void main(String[] args) {  
5   println("Jenkins Continuous Integration demo is running  
6   println("2 + 3 = " + add(2, 3));  
7  
8  
9   nt add(int a, int b) {  
10    b;  
11  
12  
13
```

View 1 edited file

build.gradle Alt+L >

... powershell - ci-jenkins-multi-build-demo + v @ [] [] ... | [] x

- PS C:\Projects\Devops\ci-jenkins-multi-build-demo> git commit -m "Update application message"
[main 73e35d9] Update application message
1 file changed, 1 insertion(+), 1 deletion(-)
- PS C:\Projects\Devops\ci-jenkins-multi-build-demo> git push
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (9/9), 604 bytes | 201.00 KiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/StoneCollector/ci-jenkins-multi-build-demo.git
651845a..73e35d9 main -> main
- PS C:\Projects\Devops\ci-jenkins-multi-build-demo>

The screenshot shows the Jenkins interface for the 'CI-Maven-Build' job. The 'Git Polling Log' tab is selected, displaying the following log content:

```

Started on 09-Aug-2026, 12:09:00 am
Using strategy: Default
[poll] Last Built Revision: Revision 651845addaf62b08d1a1e01754304ba3dc518f (refs/remotes/origin/main)
The recommended git tool is: NONE
No credentials specified
> C:\Program Files\Git\cmd\git.exe --version # timeout=10
> git --version # 'git version 2.45.1.windows.1'
> C:\Program Files\Git\cmd\git.exe ls-remote -h -- https://github.com/StoneCollector/ci-jenkins-multi-build-demo.git # timeout=10
Found 1 remote heads on https://github.com/StoneCollector/ci-jenkins-multi-build-demo.git
[poll] Latest remote head revision on refs/heads/main is: 73e35d9e273488d9b8bec296fccbf8f935c772ab
Done, Took 1.3 sec
Changes found
  
```

On the left sidebar, the 'Builds' section shows a list of builds for 'Today' and 'August 8, 2026':

- Today: #2 12:09 AM
- August 8, 2026: #1 11:57 PM

Changes detected in git
Build failure test

The screenshot shows a code editor with the following Java code:

```

1 package com.vit.demo;
2
3 import org.junit.Test;
4 import static org.junit.Assert.fail;
5
6 public class AppTest {
7     @Test
8     public void testAdd() {
9         int expected = 6;
10        int actual = App.add(2, 4);
11        if (expected == actual) {
12            System.out.println("TEST PASSED: App.add(2, 4)");
13        } else {
14            System.out.println("Test failed: expected " + expected + " but actual was " + actual);
15            fail("Test failed: expected " + expected + " but actual was " + actual);
16        }
17    }
18
19 }
  
```

A tooltip 'Chat Ctrl+L ...' is visible over the code. At the bottom, there are buttons for 'View 1 edited file' and 'build.gradle Alt+L >'. The line number 6 is highlighted in the original image.

Changed expected to 6 instead of 5

The screenshot shows the Jenkins interface for the 'CI-Maven-Build' job. The 'Status' tab is selected, showing a red 'X' icon and the text 'CI-Maven-Build Checks out the Java application'. Below this, there is a section for 'Last Successful Artifacts' showing 'ci-jenkins-demo-1.0.0.jar' (2.61 KB) with a 'view' link. The 'Permalinks' section lists the following build history:

- Last build (#3), 57 sec ago
- Last stable build (#2), 4 min 58 sec ago
- Last successful build (#2), 4 min 58 sec ago
- Last failed build (#3), 57 sec ago
- Last unsuccessful build (#3), 57 sec ago
- Last completed build (#3), 57 sec ago

On the left sidebar, the 'Builds' section shows a list of builds for 'Today' and 'August 8, 2026':

- Today: #3 12:13 AM, #2 12:09 AM
- August 8, 2026: #1 11:57 PM

Conclusion:	As shown above, I have tested the working of Jenkins with Maven, Ant and Gradle.
-------------	--